

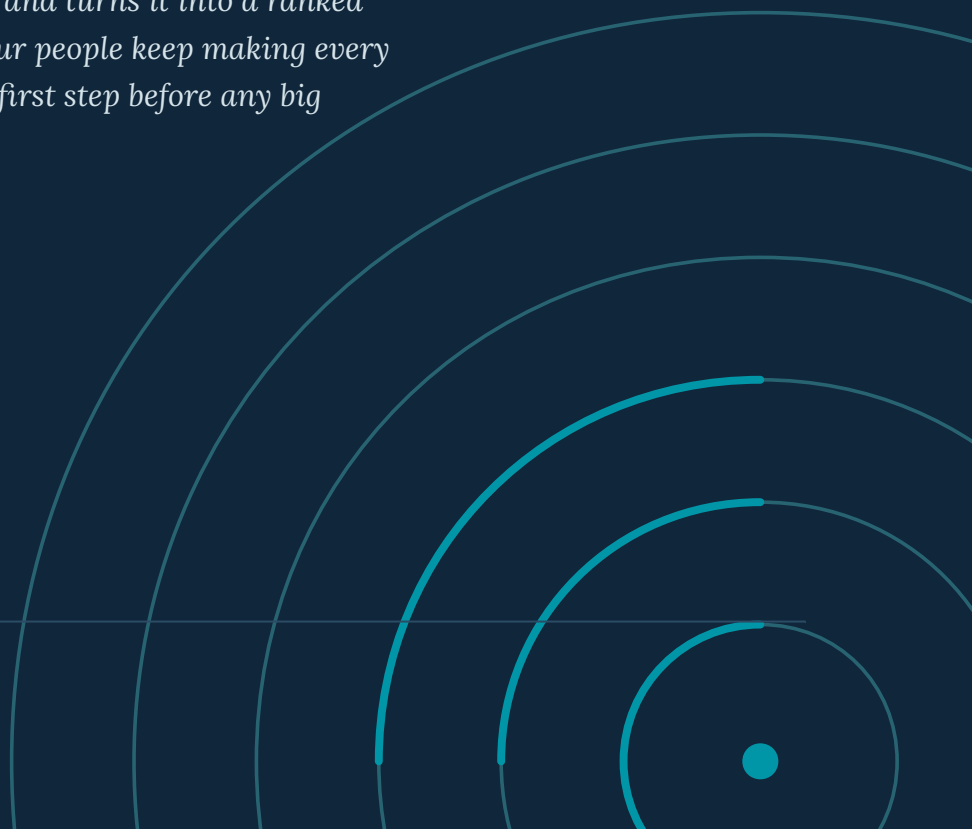
EXECUTIVE MANAGEMENT BRIEFING

SGW Resilience Platform

For senior leadership and decision-makers. Five short sections: value, money, plan, safety, growth. One page each, written to be read without any technical background — the technology lives in a separate document, the PRD.

The proposal in one sentence: *one platform that gathers SGW's scattered storm information and turns it into a ranked list of risks with plain reasons — your people keep making every decision — built as a small, testable first step before any big spend.*

FIVE SECTIONS · ONE PAGE EACH





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Five short sections: value, money, plan, safety, growth — one page each.



SGW serves more than eight million people with power and water, in a region hit by hurricanes, floods, heatwaves and wildfires. When a storm approaches, your teams make hard decisions well — but the facts they need are scattered across four computer systems that do not talk to each other. So the storm plan is built by hand over many hours, the damage picture is assembled from radio calls and a whiteboard, and when a regulator phones, there are no numbers ready. Every hour lost this way has a price: longer outages, penalty payments, expensive emergency crews, rising insurance costs.

We propose one platform that fixes exactly this — and nothing more. It gathers the scattered facts into one place and shows, for every asset, how much danger it is in and why, in plain words. Your people keep making every decision; the computer only informs them. And because the idea rests on guesses that deserve proof — above all, that seeing everything in one place really changes decisions — we do not ask for a big program. We ask for a few weeks of looking at the real work, and one small testable version. Money follows proof.

The five sections ahead: first, what actually changes for your people and your company (value). Second, where the money goes and how it comes back (finance). Third, the plan, built so that every phase must earn the next (roadmap). Fourth, who is responsible and what keeps the AI safe (governance). Fifth, how the same platform grows to new regions and new infrastructure (scalability).

#	Section	In one line
1	Strategic Business Value	Planning in minutes instead of hours; repairs ordered by facts; numbers always ready.
2	Financial Implications & ROI	The big cost is connecting old systems, not AI. No invented return numbers — money moves step by step.
3	Delivery Roadmap	Four phases; each must show written proof before the next gets budget.
4	Governance & Compliance	A named owner, built-in protections, an off-switch — the AI never acts alone.
5	Scalability	One proven pattern, repeated: new regions, new infrastructure, new workflows.



This section answers: *What actually improves, and why it is worth doing.*

Picture the day before a hurricane. Today, your operations manager spends hours collecting weather emails, exporting maps, and phoning the maintenance team — building the crew plan by hand. When the forecast shifts in the evening, that work starts again from zero. With the platform, the manager opens one screen and sees every asset ranked by risk, each with a plain reason: “old transformer, flood zone, strong wind expected.” When the forecast shifts, the list re-ranks itself in minutes. Planning stops being a race against the collection of facts and becomes what it should be: judgment.

Now picture the night of the storm. Today, your dispatcher builds the damage picture in their head from radio calls, alarms and a whiteboard — about half an hour for one full picture, while new damage reports keep arriving. With the platform, damage and repairs live on one board that updates in seconds, and the most critical facilities — hospitals, water plants — stand at the top of the repair list because facts put them there, not the loudest voice on the radio.

And picture the morning after, when a regulator or a journalist calls. Today there are no numbers ready — only phone calls to the control room. With the platform, a current, time-stamped summary — outages, crews, restoration estimates — exists at every moment. The conversation changes from “we believe we responded well” to “here is the record.”

Why this reaches the company's four pressures. Rising costs, more disruptions, growing insurance premiums, regulatory pressure — all four share one root: slow, blind storm decisions. Shorter outages mean fewer penalties and less spending on emergency help. Provable decisions change SGW's standing with regulators and give insurers a reason to reconsider premiums. One honest caution belongs here: this value rests on guesses — above all that a combined view really changes decisions, and that operators will act on computer rankings. The first small version exists to prove exactly that before big money is spent.

Who	The change in one line
Operations manager	Storm planning: from hours of collecting to minutes of judging.
Dispatcher	One live damage board instead of radio, alarms and a whiteboard.
Executive	Real numbers, time-stamped, ready at any moment.

Bottom line: *faster, fact-based storm decisions attack all four business pressures at their shared root — and the claim is tested before it is trusted.*



This section answers: *Where the money goes, how it comes back, and what the return could look like.*

Where the money goes. The largest cost is not the AI — it is the plumbing. Most of the budget goes into connecting the four existing systems: pulling data out of older software, matching records that use different codes for the same asset, and building backup paths where no modern connection exists. The second cost is running the platform: hosting, support, and keeping the AI healthy by checking its predictions against reality after every storm. The third is change itself: training, and one storm season of working the new way alongside the old way, until trust is earned.

How the money comes back. Three ways. Shorter outages — because crews wait closer to where damage will happen — mean fewer penalty payments and faster restoration. Better placement means fewer wasted trips and less premium-priced emergency help from neighboring utilities. And provable, recorded decisions strengthen SGW's position with regulators and insurers — a quieter benefit, but over years possibly the largest one.

A light simulation — what the return could look like. The numbers below are illustrations, not promises: the real baseline is measured in the first weeks of discovery. Suppose a bad storm season costs SGW \$20 million today in penalties, emergency crews, overtime and regulatory findings. Suppose the platform costs \$2.5 million to build and \$1 million per year to run — about \$5.5 million over three years. Then the question becomes: by how much can fact-based decisions cut those storm losses?

Scenario	Storm losses cut by	Saved over 3 years	Platform cost, 3 years	Result (ROI)
Cautious	5%	\$3.0M	\$5.5M	Does not pay — below breakeven (-45%)
Moderate	10%	\$6.0M	\$5.5M	About breakeven (+9%)
Strong	20%	\$12.0M	\$5.5M	Clearly positive (+118%)

What this simulation really says. On these illustration numbers, the platform pays for itself if storm losses drop by roughly 9%. So the whole investment reduces to one measurable question: can seeing the facts cut storm losses by about one-tenth? That is exactly what the first version and the shadow season are built to measure — before full commitment. Two more honest notes: the result is very sensitive to the baseline (a \$40M storm season halves the required improvement to ~5%), and because one missed major failure costs roughly a thousand times more than a false alarm, a single caught disaster can dominate the whole calculation on its own.

Bottom line: *we will not sign an invented ROI — but on any reasonable numbers, the breakeven bar is low (cut storm losses by about a tenth), and the first phases exist to measure whether SGW clears it.*



This section answers: *The plan, its milestones, and what could block it.*

The plan has four phases, and one rule connects them: each phase must produce written proof before the next phase receives budget. Nothing starts on hope.

Phase 0 — Discovery (weeks). We sit in the real control room and watch real storm work — because plans built on guessed workflows fail quietly. In the same weeks we check the two facts everything depends on: can the data actually be pulled out of the four systems, and does a usable history of past failures exist? We also measure the baseline: what a storm costs SGW today. The phase ends with data confirmed reachable — or backup paths chosen — and the baseline on paper.

Phase 1 — The first version (months). One storm scenario, end to end, on prepared data: the joined view of all assets, the ranked risks with plain reasons, the planning screen and the dispatch board. Real operators test it hands-on. The exit test is unforgiving: operators must change a real decision in the tests and act on the rankings. If they do not, we redesign — we do not scale a product nobody's decisions depend on.

Phase 2 — Make it real (one storm season). Live connections to all four systems, and the platform runs through a real storm season alongside today's way of working — informing, not yet depended on. The season is the exam: predictions checked against what actually failed, false alarms staying cheap to dismiss, and operators keeping the habit without being told.

Phase 3 — Extend. Water early-warning, automatic summaries with human approval, the crew app, smarter models. Each extension must first name the decision it improves — the same discipline that kept version one small. The two dependencies that matter most are both checked in Phase 0, before any large commitment: access to SGW's data (backup paths ready if connections are missing) and SGW's own records of past failures — the one thing no outside data can replace.

Phase	Time	The proof that unlocks the next
0 · Discovery	Weeks	Data reachable; baseline measured
1 · First version	Months	Operators change decisions and act on rankings
2 · Make it real	A storm season	Predictions verified against reality
3 · Extend	Ongoing	Each addition names the decision it improves

Bottom line: *weeks to know the facts, months to prove the idea, one season to trust it — and budget follows proof at every step.*



This section answers: *Who is responsible, and what keeps the AI safe and lawful.*

Start with the rule that governs everything: the AI recommends — your people decide. The system cannot dispatch a crew, close a valve, or switch anything, anywhere, by itself. Technically it can only read from the systems that control the grid and water; the connection is built one-way. In the worst imaginable case — an attacker fully taking over the platform — they would see information, but they could not touch the physical network. The platform is a window, never a remote control.

Responsibility has a name. The operations director — a person, not a department, and not the software vendor — owns what the AI recommends and its consequences. Once a month, operations, engineering and compliance sit at one table and review how the AI performed: what it predicted, what actually happened, what operators overrode and why.

We planned for what could go wrong — in writing, before launch. Four dangers, each with a protection built into the design. Help arriving late at a hospital: impossible for the AI to cause alone, because a person confirms every crew movement. Poorer neighborhoods quietly receiving less protection than richer ones: the rankings are tested for fairness across areas and income groups, before launch and continuously after. People trusting the computer too much and switching off their own judgment: every ranking shows its reasons in plain words, and any alarm can be dismissed in one click — the design keeps humans thinking. A wrong number reaching a regulator or the press: no summary leaves the building without a person approving it.

And if leadership ever wants out: there is an off-switch. At any moment SGW can turn the AI off and keep working manually, with all data still visible. On the legal side, three checks form a named early task with your legal team: the US security rules for power companies (called NERC CIP) — our one-way design aims to keep the platform outside the most heavily regulated zone, to be verified rather than assumed; the state regulator's reporting duties — which the platform makes easier, since numbers and records are always ready; and privacy — screens show neighborhood totals rather than single homes wherever possible. The whole approach follows the US government's published framework for trustworthy AI (NIST, 2023).

The danger	The built-in protection
Late help to a hospital	A person confirms every crew movement
Unfair protection between neighborhoods	Rankings tested for fairness, continuously
Over-trust in the computer	Reasons shown on every ranking; one-click dismissal
Wrong number reaching a regulator	Human approval before any summary leaves

Bottom line: *every recommendation carries a human decision on top, a named owner behind it, and a written record underneath.*

This section answers: *How the same platform grows — regions, infrastructure, workflows.*

Version one is small on purpose — but it is built as a pattern that repeats. Growing to a new region does not mean a new project: it means connecting that region's data sources while the platform, the screens, the governance and the human-approval step stay exactly the same. Coastal regions stress hurricane and flood risk; inland regions stress heat and wildfire — so the risk models are re-tuned per region, on the same foundation.

Growing to new types of infrastructure follows the same logic. Water comes first — its sensors largely exist already, and the danger of a missed pump failure is direct: pressure loss, boil-water advisories, public health. The method transfers by re-asking one question per asset class: what does each kind of mistake cost here? For water, a missed failure carries a different price than for electricity — so the warnings are re-tuned to that price, while the five-layer platform, the one-way safety wall and the audit trail are reused unchanged. Gas and telecom can follow the same path.

Growing to new workflows is guarded by the same discipline that kept version one lean. A crew phone app, photo capture from the field, route suggestions — each is welcome only after it names the decision it improves and passes the same six-question test every AI ability passed. That admission rule is not bureaucracy; it is how the product avoids becoming a bloated dashboard nobody uses.

The order of growth is deliberate. Electricity first, where the pain and the data are clearest. Water second, where sensors exist and public health is on the line. Everything else earns its place. And one principle rules every expansion: scale only what has passed its test — never scale a guess.

Direction of growth	What changes	What stays the same
New regions	Connect that region's data; re-tune risk models	Platform, screens, governance
New infrastructure (water, then gas, telecom)	Re-price the mistakes per asset class	The five layers, safety wall, audit trail
New workflows (crew app, photos, routes)	Each names the decision it improves first	The admission rule itself

Bottom line: *growth is repetition of a proven pattern — every expansion inherits the proof requirements, the human control, and the audit trail of version one.*